Report Of **Evaluation of AI Software Developer Intern at Trovex.ai**

Submitted To
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE (AI/ML)

By

Name of Student: YASH MAHINDROO
Roll Number: CS-2341159
Section: 4CSE-1
Batch: 2023-2027

Organization: Trovex.ai
Role: AI Software Developer Intern

2026-27

CANDIDATE'S DECLARATION

I, Yash Mahindroo, do hereby declare that the internship report titled "Report Of Evaluation of AI Software Developer Intern at Trovex.ai" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Student Name: Yash Mahindroo

Roll No.: CS-2341159

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the internship program at Trovex.ai. I am thankful for their aspiring guidance, invaluable constructive criticism, and friendly advice during this work. I am sincerely grateful to my mentors and colleagues at Trovex.ai for sharing their truthful and illuminating views on a number of technical issues related to this work, and to the faculty of the School of Computer Science and Engineering, IILM University, for their continuous support and guidance throughout the internship.

I also express my gratitude to my parents for their blessings.

Student Name: Yash Mahindroo

Roll No.: CS-2341159

TABLE OF CONTENTS

4. Project Description

4.1 Introduction

This report presents the work carried out during my internship as an AI Software Developer at Trovex.ai. The internship focused primarily on two areas of the company's AI infrastructure: improving the quality of a Retrieval-Augmented Generation (RAG) pipeline built on a Neo4j knowledge graph, and evaluating supporting tools and processes for the quality assurance (QA) function of the team. Over the course of the internship, I worked on benchmarking reranking models for retrieval quality, evaluating a no-code QA automation tool against the team's existing manual testing workflow, contributing to the governance of prompt assets used across the company's LLM-based features, and supporting manual regression testing on the company's product.

4.2 Organization Profile

Trovex.ai is an AI-powered sales enablement and training company, founded in 2023 and headquartered in Bengaluru, India. Its platform uses AI-driven roleplay simulations to help sales and customer support teams practice real-world conversations, receive personalized feedback, and improve their pitching, objection-handling, and negotiation skills. The company's engineering workflow makes extensive use of large language models, both for internal development processes (such as automated test case generation) and as part of the product's simulation and feedback features.

4.3 Problem Statement

As Trovex.ai's knowledge graph and RAG pipeline scaled, the team needed a data-driven way to evaluate whether its retrieval quality could be improved through the addition of a reranking stage, and if so, which reranking approach (a managed cloud service versus open-source models) offered the best balance of accuracy and operational cost. In parallel, the QA process for validating new product features relied entirely on manual test execution, which the team wanted to evaluate for potential automation. There was also a need to bring structure and consistency to how prompt assets were organized and named within the team's LLM observability platform, as the absence of a naming convention was making prompts difficult to track and maintain as the number of features grew.

4.4 Internship Objectives

- Evaluate and benchmark a managed cloud-based reranking service against established open-source reranking models to determine which offers the best retrieval accuracy for the company's knowledge graph.
- Diagnose and document data quality issues within the production knowledge graph that could affect the validity of any retrieval or reranking evaluation.
- Assess the feasibility of an AI-driven, no-code QA automation tool as a potential replacement or supplement to the team's manual testing workflow.
- Establish a naming convention and folder structure for prompt assets within the team's LLM observability platform to improve maintainability.
- Support the team's manual QA process by testing new features on the staging environment prior to production release.

4.5 Scope of the Internship

The scope of this internship was limited to the evaluation, testing, and infrastructure-support layer of Trovex.ai's engineering workflow, rather than core product feature development. Specifically, the work covered: (a) building an evaluation pipeline to compare reranking approaches for the existing RAG system without modifying the production retrieval pipeline itself; (b) researching and trialing a third-party QA automation tool for feasibility assessment, stopping short of full production adoption pending management approval; (c) organizing and standardizing prompt management practices within the existing observability tooling; and (d) manual functional testing of new features on the staging/beta environment prior to production promotion. Production deployment decisions and final tool adoption remained outside the scope of the internship and were left to the team's management.

4.6 Technologies and Tools Used

- Neo4j: A graph database used to store the company's knowledge graph, including extracted entities, document metadata, and vector embeddings. Used throughout for schema inspection, Cypher-based querying, and vector similarity search via its native vector index.
- Python: The primary language used to build the evaluation pipelines, using the official Neo4j driver for graph queries, Google Cloud client libraries for

embedding and reranking calls, and the sentence-transformers and FlagEmbedding libraries to run open-source reranker models locally.

- Google Cloud Platform (Vertex AI): Used for two purposes — generating text embeddings (via the gemini-embedding-001 model) for semantic vector search, and accessing the managed Vertex AI Ranking API (Discovery Engine) as one of the reranking approaches under evaluation.
- Open-source reranking models (BGE-reranker-v2-m3, MiniLM-L6-v2): Cross-encoder models run locally to provide a baseline comparison against the managed cloud reranking service, evaluated on both accuracy and latency.
- Momentic.ai: A no-code, AI-driven QA automation tool evaluated as a potential replacement for parts of the team's manual web application testing workflow, capable of generating and executing browser-based test cases against a staging deployment.
- Langfuse: An LLM observability and prompt management platform already in use by the team for pipeline tracing and evaluation. As part of this internship, ownership of the prompt folder structure was taken on, including renaming existing prompts and establishing a consistent naming convention for future prompt assets.
- NotebookLM: Used to generate realistic, document-grounded evaluation questions from the company's source documents, which served as the query set for the reranker evaluation.

4.7 System Architecture

The core retrieval architecture evaluated during this internship follows a two-stage retrieval-and-rerank design, common in production RAG systems:

- User query is converted into a vector embedding using Vertex AI's embedding model.
- The embedding is used to perform an approximate nearest-neighbour vector search against Neo4j's vector index, scoped to the relevant document folder, returning a shortlist of candidate entities.

- The candidate shortlist is passed to a reranking stage — either the managed Vertex AI Ranking API or a locally-run cross-encoder model (BGE or MiniLM) — which re-scores each candidate against the query for finer-grained relevance.
- The reranked results are returned as the final ranked output, from which the top result(s) would be used to answer the user's query.

Separately, the QA automation evaluation followed a browser-automation architecture, where Momentic.ai drives a browser against the team's staging/beta deployment to execute generated test cases, in comparison to the team's existing workflow of manually executing test cases generated with the help of an LLM.

4.8 Methodology

The reranker evaluation was designed as a controlled comparison: the same set of evaluation questions and the same retrieved candidate pool were used across all three reranking approaches, so that the reranking stage was the only variable being measured. Ground truth was established by manually mapping each evaluation question to its correct source document. Each reranker's output was then scored using standard information retrieval metrics — Accuracy@1 (whether the correct result was ranked first) and Accuracy@3 (whether it appeared within the top three results) — along with average response time per query.

Before running this evaluation, the underlying knowledge graph was audited for data quality issues, including duplicate document ingestion and inconsistent folder-based data organization, which could otherwise have invalidated the comparison. A clean, single-copy dataset was established prior to running the final evaluation.

The QA automation tool was evaluated through a hands-on proof-of-concept setup, with adoption paused pending management approval and resolution of a data security consideration around providing staging-site credentials to a third-party tool. Manual testing of new features continued in parallel using the team's existing process of generating test cases with the help of an LLM and executing them by hand on the staging environment before promotion to production. For the prompt management work, existing prompts within the observability platform were reviewed, renamed, and reorganized according to a newly proposed folder and naming convention intended to make prompt assets easier to locate and maintain as the platform grows.

4.9 Expected Outcomes

- A data-driven recommendation on reranking approach for the company's RAG pipeline, based on measured accuracy and latency differences between the managed cloud service and open-source alternatives.
- A documented set of data quality issues in the production knowledge graph, providing the team a basis for improving future data ingestion practices.
- A feasibility assessment of the QA automation tool to support management's decision on whether to adopt it for the team's testing workflow.
- An improved, standardized naming and folder convention for prompt assets, reducing ambiguity and maintenance overhead for the team going forward.
- Continued QA coverage for new feature releases through manual testing support during the internship period.

4.10 Offer Letter and Other Communication Proof

Le#er of Offer

Dated : 01/06/2026

Dear Yash

We are pleased to offer you a Full **time internship- AI Software Development** with **Cloudsprint Technologies** for a tenure of three months starting 3rd June 2026.

a) During your employment, you will report directly to **Dhruv Shindhe, Co-Founder**. You will be expected to carry out the dues assigned to you in a competent and efficient fashion.

b) This is an unpaid internship designed to provide practical exposure to software development in a fast-paced startup environment. While no spend or salary will be provided, the internship offers hands-on project experience, mentorship, and an opportunity to work closely with the founding and product teams.

c) Your place of posting will be Remote.

d) The normal working hours are 9.30 A.M. to 6.30 P.M. with a 30 minutes lunch break Monday through Friday. However due to business exigency you may be required to work at different timings which may be prescribed by the manager from time to time.

e) You may be required to work at different times including night. In all such cases, the actual working hours shall be as prescribed by the manager.

f) Termination of internship will be immediate effect by either party on confirmation.

g) If your actions at any time constitute a serious breach of Cloudsprint's standards of behaviour, we may end this contract and terminate your employment immediately.

Yours sincerely,
For Cloudsprint Technologies Private Limited,
Amaresh Ojha
Chief Executive Officer

Address: NO 19 4TH C CROSS, INDUSTRIAL AREA 5TH BLK,Koramangala
Bangalore South,Bangalore-560095,Karnataka

5. Bibliography / References

Neo4j Documentation. Neo4j Graph Database and Vector Index. Available at: https://neo4j.com/docs/

Google Cloud. Vertex AI Documentation — Text Embeddings and Ranking API. Available at: https://cloud.google.com/vertex-ai/docs

Hugging Face Model Hub. BAAI/bge-reranker-v2-m3 Model Card. Available at: https://huggingface.co/BAAI/bge-reranker-v2-m3

Hugging Face Model Hub. cross-encoder/ms-marco-MiniLM-L6-v2 Model Card. Available at: https://huggingface.co/cross-encoder/ms-marco-MiniLM-L6-v2

Momentic.ai Documentation. AI-Driven QA Automation Platform. Available at: https://momentic.ai/

Langfuse Documentation. LLM Observability and Prompt Management. Available at: https://langfuse.com/docs

Google NotebookLM. Available at: https://notebooklm.google/